Garry Nelson

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Electrical Engineering fresh graduate (Summa Cum Laude, GPA 3.98/4.00) from Telkom University with a specialization in Machine Learning and Deep Learning for real-world sensing systems. Published author in IEEE Access (Q1) with research focused on real-time human fall detection using FMCW radar and deep learning architectures. Experienced in building end-to-end AI systems spanning model design, training, and real-time inference, as well as IoT and embedded systems development. Seeking ML Engineer or Deep Learning Researcher roles where research depth and applied engineering can drive real-world impact.

Education

Telkom University – Bandung, Indonesia (September 2022 – January 2026)

Bachelor of Electrical Engineering – GPA: 3.98 / 4.00 – Summa Cum Laude

Relevant coursework: Machine Learning, Artificial Intelligence, Digital Signal Processing, Linear Algebra, Probability & Statistics, IoT Systems

Projects

Real-Time Human Fall Detection System using Radar and Deep Learning

(https://github.com/notgarryy/real-time-fall-detection-system-with-TTSNet)

- Designed and trained a TTSNet-based deep learning model for spatiotemporal feature extraction on FMCW radar point cloud data, achieving 98% classification accuracy across multiple human activity classes
- Implemented a real-time inference pipeline and processing live radar data for immediate fall detection response
- Research extended into a TCN-based architecture variant, resulting in two peer-reviewed publications (IEEE Access Q1 + Q3 SINTA 1)

Fall Detection Inference API

(https://github.com/notgarryy/fall-detection-api)

- Built a production-ready REST API using FastAPI that accepts FMCW radar point cloud data and returns real-time fall detection results with confidence scores
- Implemented full preprocessing pipeline including DBSCAN clustering, sliding window inference, and TTSNet model serving
- Containerized with Docker and deployed on cloud, demonstrating end-to-end MLOps capability

Face Recognition-Based Smart Door Lock System

(*https://github.com/notgarryy/facenet-face-detection*)

- Implemented a face authentication system using FaceNet for feature extraction and identity matching, achieving 88.9% recognition accuracy
- Deployed on Raspberry Pi 4 for real-time processing and physical door lock actuation
- Addressed lighting inconsistency through targeted data augmentation, improving model robustness across varying environmental conditions

Publications

1. **Spatiotemporal Feature Learning for Real-Time Human Fall Detection System Using TCNs and FMCW Radar Point Clouds** – IEEE Access (Q1), 2026 **|** DOI: *doi.org/10.1109/ACCESS.2026.3676850*
2. **Real-Time Fall Detection Using FMCW Radar Point Clouds with TTSNet Spatiotemporal Feature Learning –** Advance Sustainable Science, Engineering and Technology (Q3, SINTA 1), 2026 **|** Accepted (in revision)
3. **Enhancing STEM Education through IoT-Based Distance Monitoring Projects in High Schools –** Smart Society (SINTA 2), 2026 **|** DOI: *doi.org/10.58524/smartsociety.v6i1.1063*

4. **Pelatihan Internet of Things Berbasis STEM untuk Pengembangan Kompetensi Digital Siswa dan Guru SMA** – Jurnal Masyarakat Mandiri (SINTA 3), 2025 **|** DOI: *doi.org/10.31764/jmm.v9i4.32947*

Work Experience

PT. Labtech Penta International Batam – Batam, Kepulauan Riau

Digital Twin Intern (June 2025 – August 2025)

- Developed a Unity-based Digital Twin interface for an HVAC training system, enabling real-time monitoring and control of physical equipment states through a synchronized virtual environment
- Designed and integrated an MQTT broker on Google Cloud Platform to handle bidirectional communication between the physical HVAC trainer and the Digital Twin application, transmitting live state and control logic data
- Collaborated within a 6-person team

Organizational Experience

Sensing, Signal Processing and Learning for Spatial Intelligence Laboratory – Telkom University

Research Assistant (July 2025 – January 2026)

- Conducted research on Human Activity Recognition (HAR) using FMCW radar point cloud data
- Developed and evaluated deep learning models for spatiotemporal feature extraction and real-time activity classification, contributing directly to two peer-reviewed publications

Basic Computing Laboratory – Telkom University

Teaching Assistant (July 2023 – June 2025)

- Assisted approximately 3 students per session across 20 sessions per semester in C programming fundamentals and debugging
- Mentored 5 students per semester on final projects, with mentees consistently achieving above-average outcomes relative to their peers

Electronics and Industrial Automation Laboratory – Telkom University

Teaching Assistant (January 2024 – December 2025)

- Facilitated IoT communication labs using ESP32 for approximately 4 students per session across 8 sessions per semester
- Taught PLC programming fundamentals (Omron) and Ladder Logic to undergraduate students in industrial automation courses

Technical Skills

- Machine Learning & AI: TensorFlow, Keras, Scikit-learn, Deep Learning, Convolutional Neural Networks, Temporal Convolutional Networks (TCN), Time-Series Modeling, Human Activity Recognition
- MLOps: FastAPI, Docker, MLflow, GitHub Actions
- Data & Analytics: NumPy, Pandas, Matplotlib, Seaborn, SQL
- IoT & Embedded Systems: ESP32, Raspberry Pi, Arduino, MQTT, BLE, Sensor Integration, PLC (Omron), Ladder Logic
- Signal Processing & Sensing: FMCW Radar, Point Cloud Processing, Real-Time Inference, Feature Extraction
- Programming: Python, C, Flutter, Git